

CO5 -AT2-SHORT ANSWERS TEST

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REG NO :192525109

COURSE CODE : CSA0307

COURSE NAME : DATA STRUCTURES

Q1).

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QUESTIONS

Q1 Which traversal algorithm is commonly used to find the shortest path in an unweighted graph? 2 marks

Which traversal algorithm is commonly used to find the shortest path in an unweighted graph?

Your Answer

Breadth first search (BFS) Because it checks node level by level.

Save Answer Next →

Q2 Define Dijkstra's algorithm. 2 marks

Q3 Define a Minimum Spanning Tree (MST). 2 marks

Q4 What is the difference between a spanning tree and a minimum spanning tree? 2 marks

Q5 What is the time complexity of Prim's algorithm using an adjacency matrix? 2 marks

Q6 A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order. 2 marks

Q2).

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QUESTIONS

Q1 Which traversal algorithm is commonly used to find the shortest path in an unweighted graph? 2 marks

Q2 Define Dijkstra's algorithm. 2 marks

Define Dijkstra's algorithm.

Your Answer

It is used to find shortest path from single source to all vertices with non negative edge weight

Save Answer Next →

Q3 Define a Minimum Spanning Tree (MST). 2 marks

Q4 What is the difference between a spanning tree and a minimum spanning tree? 2 marks

Q5 What is the time complexity of Prim's algorithm using an adjacency matrix? 2 marks

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Q3).



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Q2

Define Dijkstra's algorithm.
2 marks

Q3

Define a Minimum Spanning Tree (MST).
2 marks

Q4

What is the difference between a spanning tree and a minimum spanning tree?
2 marks

Q5

What is the time complexity of Prim's algorithm using an adjacency matrix?
2 marks

Q6

A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order & critical network.

Q3 Define a Minimum Spanning Tree (MST). 2 marks

Define a Minimum Spanning Tree (MST).

Your Answer

MST is an spanning tree with minimum total edge weight

Save Answer Next →

Q4).



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A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order & critical network.

Q4 What is the difference between a spanning tree and a minimum spanning tree? 2 marks

What is the difference between a spanning tree and a minimum spanning tree?

Your Answer

A Spanning tree connects all vertices without cycle while, MST is spanning tree with minimum total edge weight

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Q5).



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2 marks

Q5 ✓
What is the time complexity of Prim's algorithm using an adjacency matrix?
2 marks

Q6 ✓
A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order. A social network represents users as ve

Q5 What is the time complexity of Prim's algorithm using an adjacency matrix? 2 marks

What is the time complexity of Prim's algorithm using an adjacency matrix?

Your Answer

Time complexity of prim's algorithm is $O(v^2)$ where v is number of vertices

Save Answer

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Q6).



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Q6 A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order. A social network represents users as ve 2 marks

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A social network represents users as ve

Your Answer

A-B-C-D-E-F

Save Answer

Next →

Q7)



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A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS

Q7 2 marks
A project contains the dependencies $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, and $S \rightarrow T$. Determine whether the tasks can be completed using topological sorting and identify one valid execution sequence.

A project contains the dependencies $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow S$, $R \rightarrow S$, and $S \rightarrow T$. Determine whether the tasks can be completed using topological sorting and identify one valid execution sequence.

Your Answer

yes, it is one valid topological order P-Q-R-S-T

Save Answer